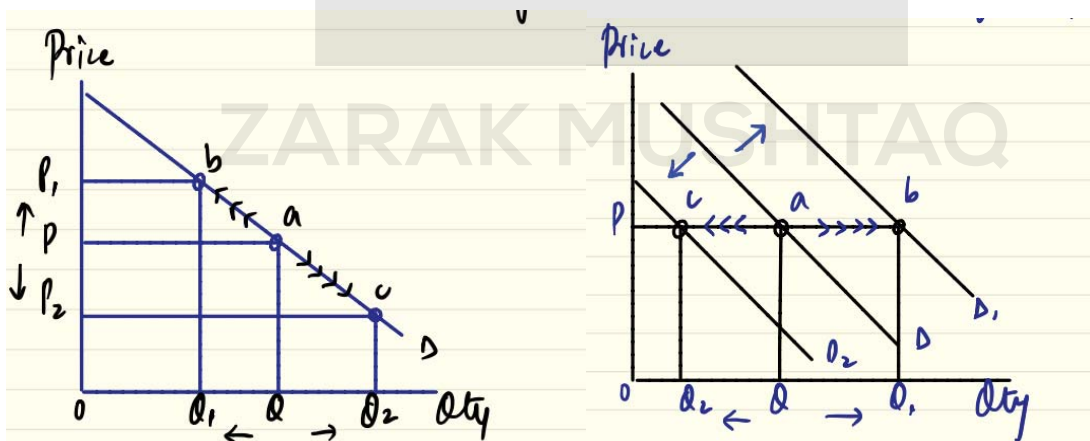


Demand & supply

Demand is a very useful tool in economics and when used in conjunction with supply can be used to predict and analyze market equilibrium. From the perspective of a producer (a business, firm, seller or supplier), demand represents the potential for sales of your products. Modern companies spend millions of dollars each year analyzing the demand for their goods and take this into account when making business decisions. Companies often try to increase the demand for their products through advertising and promotions. From the perspective of a consumer (an individual, customer, purchaser or buyer), demand represents your desire and ability to consumer products or services.

Demand represents the maximum quantity of a particular good that consumers are willing and able to buy during a specified time period. For a demand to be effective both willingness and ability must exist at the same time. The fundamental character of the concept of demand is the relationship between the price of a good and the maximum quantity that is demanded and is described as the law of demand.

Law of demand states that as the price of good increases the quantity demanded contracts and opposite is also true, as the price of good decreases the quantity demanded extends. Movement along the demand curve will only be caused by the factor price and the other factors will remain constant i.e. keeping all other factors constant (*ceteris paribus*). There are a number of factors that affect demand and several ways to represent demand.



FACTORS AFFECTING DEMAND: Along a demand curve the factors, other than the price of the good, are held constant. When these factors change the demand

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Demand & supply

curve shifts either outwards or inwards which means that either more or less is demanded at each and every price. This is called as an increase or decrease in demand.

INCOME: It represents purchasing power which can affect two types of goods. The effect that income on the amount of a product that consumers are willing and able to buy depends on the type of good. For most goods, there is a positive (direct) relationship between a consumer's income and the amount of the good that one is willing and able to buy. In other words, for these goods when income raises the demand for the product will increase: when income falls, the demand for the product will decrease. These types of goods are called normal goods.

However, for some goods the effect of a change in income is the reverse. For example, think about a low quality (high fat-content) ground beef. You might buy this while you are a student, because it is inexpensive relative to other types of meat. But if your income increases enough, you might decide to stop buying this type of meat and instead buy leaner cuts of ground beef, or even give up ground beef entirely in favor of beef tenderloin. If this were the case (that as your income went up, you were willing to buy less high-fat ground beef), there would be an inverse relationship between your income and your demand for this type of meat. We call this type of good an inferior good. There are two important things to keep in mind about inferior goods. They are not necessarily low quality goods. The term inferior (as we use it in economics) just means that there is an inverse relationship between one's income and the demand for that good. Also, whether a good is normal or inferior may be different from person to person. A product may be normal good for you, but an inferior for another person.

PRICE OF RELATED GOODS: Two goods that are typically consumed together are called complements. If the price of car goes up, the law of demand tells us that we are willing or able to buy fewer cars and we buy less quantity of car the consumption of petrol will also decrease leading to fall in the demand. We can summarize this by saying that when two goods are complements, there is an inverse relationship between the price of one good and the demand for the other good.

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On the other hand, some goods are considered to be substitutes for one another: we don't consume both of them together, but instead choose one or the other. For example, for some people coke and Pepsi are substitutes. If the price of coke increases, this may make Pepsi relatively more attractive. The law of demand tells us that fewer people will buy coke; some of these people may decide to switch to Pepsi instead, therefore increasing the amount of Pepsi that people are willing and able to buy. We summarize this by saying that when two goods are substitutes, there is a positive relationship between the price of one good and the demand for the other good.

TASTE AND PREFERENCES OF CONSUMERS: This is a less tangible item that still can have a big impact on demand. There are all kinds of things that can change one's tastes or preferences that cause people to want to buy more or less of a product. If a celebrity endorses a new product, this may increase the demand for a product. On the other hand, if a new health study comes out saying something is bad for your health; this may decrease the demand for the product. Another example is that a person may have a higher demand for an umbrella on a rainy day than on a sunny day.

CHANGES IN POPULATION: An increase in the size of the total population leads to an increase in the demand for many goods and services causing their demand curves shift outwards. In developing countries the birth rates are very high causing the young population as a larger proportion of the total population and hence the demand for the goods and services by the young population increases.

ADVERTISING: A successful advertising campaign persuades people to buy goods and services which they originally did not need or want. Many firms spend heavily on advertising to increase their revenue and profits.

CHANGES IN INTEREST RATES: Many goods are bought on credit using borrowed money, thus the demand for them may be sensitive to the rate of interest charged by the lender. An increase in the interest rate would encourage an individual to save more rather than spend, thus reducing the consumption expenditure, hence decreasing the demand for goods and services.

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INDIVIDUAL AND MARKET DEMAND: Individual demand is the quantity which an individual consumer is willing to buy at a particular price. On the other hand market demand is the total quantity of a particular commodity which all the consumers are willing to buy at a particular price, so the market demand is the horizontal summation/aggregation of all the individual demand curves.

ELASTICITIES OF DEMAND

PRICE ELASTICITY OF DEMAND: It measures the responsiveness to the percentage change in quantity demanded to the percentage changes in price of a good i.e. how the consumer responds, by making changes in the quantities which they are willing to buy, as a result of changes in the price of a commodity. The formula for calculating the coefficient of elasticity of demand is:

$$\text{PED} = \frac{\text{percentage change in quantity demanded}}{\text{Percentage change in price.}}$$

Since changes in price and quantity nearly always move in opposite directions, economists usually do not bother to put in the minus sign. We are concerned with the coefficient of elasticity of demand. E.g. if the value of PED is -2.5 then the minus sign shows the direction of change and 2.5 is the magnitude of change which means that a change in price will bring a 2.5 times bigger change in quantity demanded.

If the percentage change in the quantity demanded is greater than the percentage change in price, the demand is said to be elastic and the value of PED will be greater than one. On the other hand, if the percentage change in quantity demanded is less than the percentage change in price, the demand is said to be inelastic and the value of PED will be less than one.

CALCULATION OF PED:

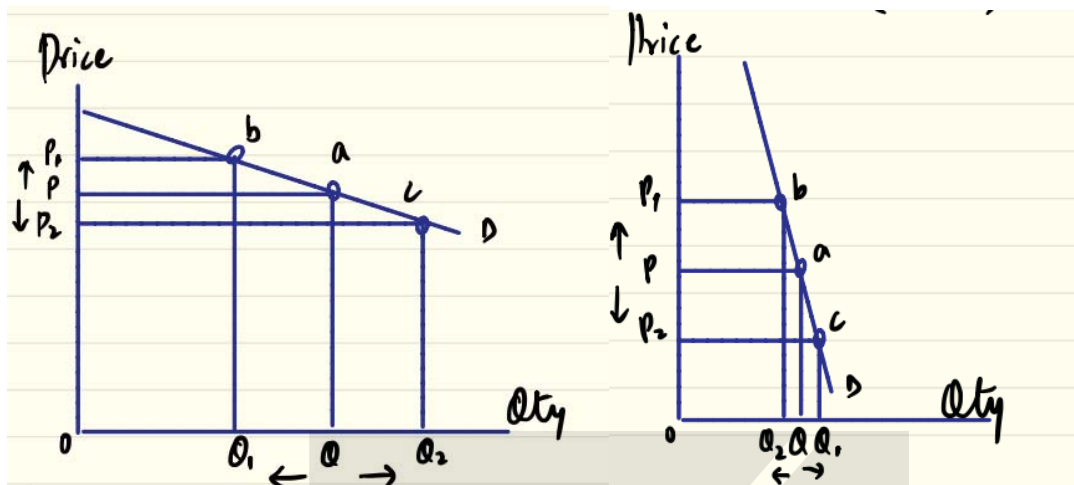
$$\% \text{ change} = (\text{new} - \text{old}) / \text{old} \times 100$$

$$\text{Percentage change in price: } (P_2 - P_1) / P_1 \times 100$$

$$\text{Percentage change in quantity: } (Q_2 - Q_1) / Q_1 \times 100$$

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WHAT DETERMINES OR INFLUENCES PED?

1. **The number of close substitutes for a good / uniqueness of the product:** The closer substitutes in the market, the more elastic is the demand for a product because consumers can more easily switch their demand if the price of one product changes relative to others in the market. The huge range of package holiday tours and destinations make this a highly competitive market in terms of pricing, many holiday makers are price sensitive.
2. **The degree of necessity or whether the good is a luxury:** Goods and services deemed by consumers to be necessities tend to have an inelastic demand whereas luxuries will tend to have more elastic demand because consumers can live without luxuries i.e. the consumption can be postponed.
3. **The percentage of a consumer's income allocated to spending on the good:** Goods and services that take up a high proportion of a household's income will tend to have a more elastic demand than products where large price changes makes little or no difference to someone's ability to purchase the product.
4. **The time period allowed following a price change:** Demand tends to be more price elastic the longer we allow consumers to respond to a price change by varying their purchasing decision. In the short-run, the demand may be inelastic, because it takes times for consumers both to notice and then to respond to price fluctuations.
5. **Whether the good is subject to habitual consumption:** When this occurs, the consumer becomes much sensitive to the price of the good in question.

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Clearly, examples such as cigarettes and alcohol and other drugs can come into this category.

6. **Peak and off-peak demand:** Demand tends to be price inelastic at peak times - a feature that suppliers can take advantage of when setting higher prices. Demand is more elastic at off-peak times, leading to lower prices for consumers. Consider for example the charges made by car rental firms during the course of a week, or the cheaper deals available at hotels at weekends and away from the high season. Train fares are also higher on Fridays (a peak day for travelling between cities) and also at peak times during the day.

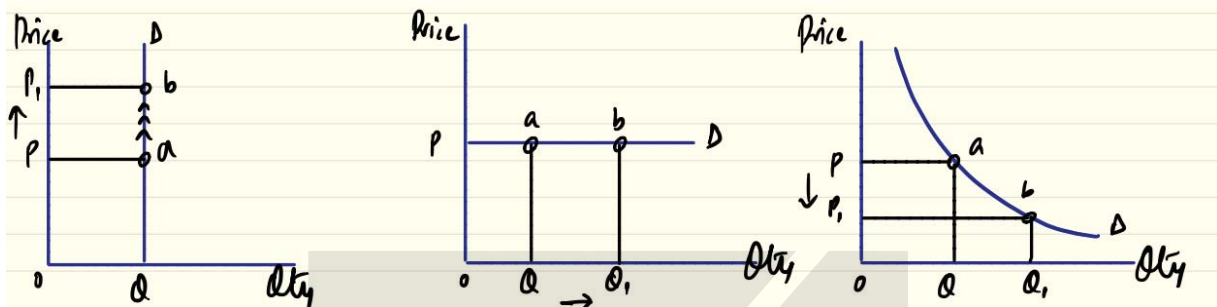
EXTERME CASES OF PED:

1. **PERFECTLY ELASTIC DEMAND:** In this situation a minute or fractional change in price brings very large or infinite changes in quantity demanded and the coefficient of PED is equal to infinity. In perfectly competitive market where there are a large number of producers selling a particular homogenous commodity at a particular market price faces a perfectly elastic (horizontal) demand curve. If one of the producers increase the price of his product a little bit the demand for his product will fall to zero i.e. no consumer will buy from him and if he decreases the price a little bit then the demand for his product will increase infinitely.
2. **PERFECTLY INELASTIC DEMAND:** In this situation a change in price brings no change in quantity demanded and the value of PED is equal to zero. There are very few examples of perfectly inelastic demand where a consumer buys the similar quantity at any price e.g. life saving drugs, blood. In this situation a change in the price of the good brings an equal percentage change in the revenue of the producer or the expenditure of the consumer.
3. **UNITARY ELASTIC DEMAND:** In this situation a change in price brings an equal percentage change in quantity demanded and the value of PED is one. In this case the total revenue of the producer or the expenditure of the consumer remains unchanged as a result of changes in the price of the good. There is no particular good which always show unitary elastic demand, it happens when a consumer fixes his expenditures on the purchase of a good.

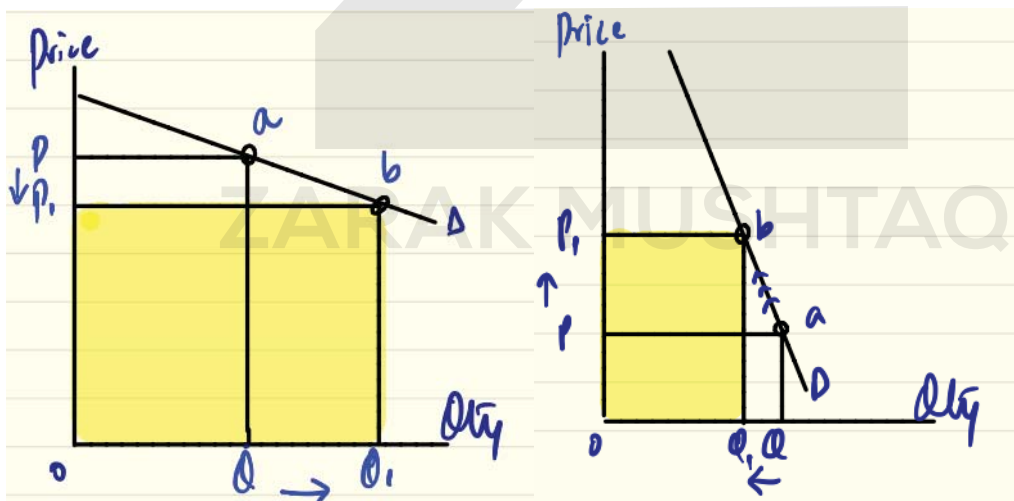
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The shape of a unitary elastic demand curve is called rectangular hyperbola which shows that the area of the percentage change in price and the area of the percentage change in quantity demanded are always equal.



Price elasticity of demand and the total revenue of producer or total expenditure of consumer: The revenue (income) of the producer depends very much on the elasticity of demand for the good which he is selling. With the knowledge of PED for his good he can make appropriate changes in the price of the good to increase his total revenue. In case of product being elastic, a change in price brings a larger proportionate change in quantity demanded, so producer should decrease the price to increase total revenue of the firm. While in case of inelastic product, a change in price brings a lesser proportionate change in quantity demand, so producer can increase price to increase its revenue.



PED varies along the demand curve: PED becomes more elastic as the price of a product rises. Consumers become more sensitive to price changes, the higher the price of the product. This is because, for example, a 10% rise in price when price

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was initially \$10 000 would involve consumers having to spend considerably more (\$1000) to buy the product. If a supplier was foolish enough to keep raising the price, a point would come when the product would be priced out of the market. At this point, demand would be perfectly elastic.

As the price falls, demand becomes more inelastic. For example, a 10% fall in price when the price was initially \$1 is not very significant and is unlikely to result in much extra demand. If the price falls to zero, there will be a limit to the amount people want to consume. At this point, demand is perfectly inelastic. Figure 11.6 shows how PED varies over a straight line demand curve. At the mid-point there is unit PED, with the percentage change in quantity demanded matching the percentage change in price.

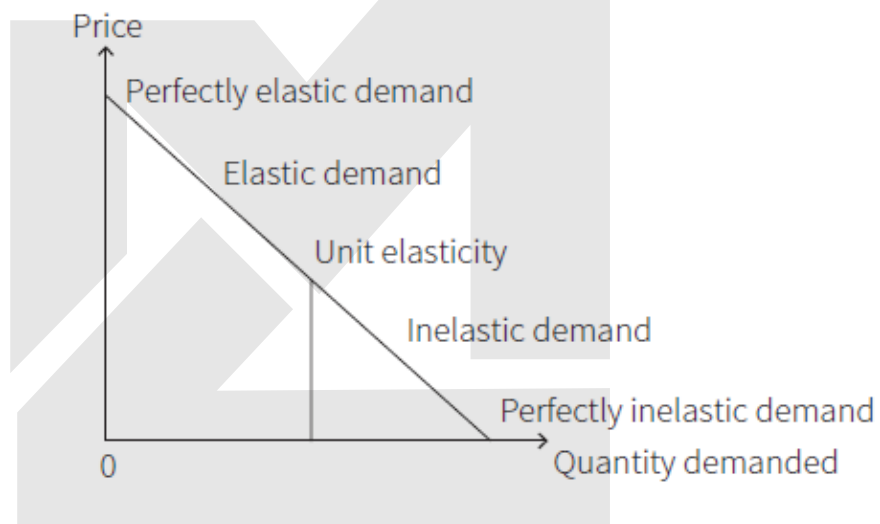


Fig. 11.6: Variation of PED over a demand curve

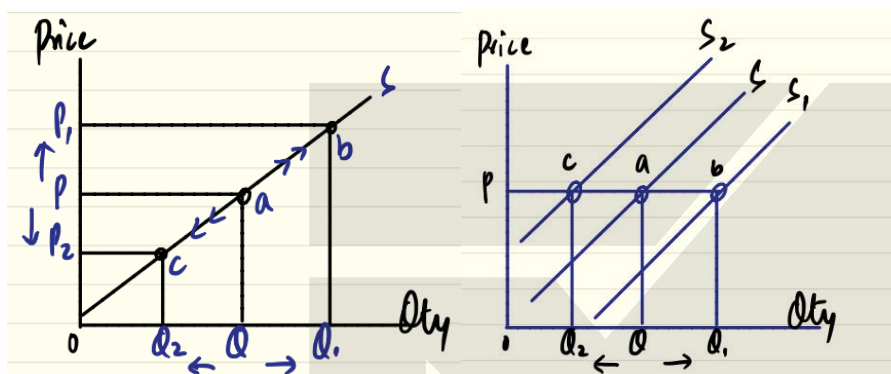
SUPPLY: Supply is a very useful tool and when used in conjunction with demand can be used to predict and analyze market equilibrium. Supply reflects the availability of products that you can purchase. Modern companies spend lots of time each year analyzing the supply for goods that must be purchased for use in their business as well as their competitors supply and take this in to account when making business decisions. From the perspective of a producer (a business, firm, seller or supplier), supply reflects your desire and ability to provide products or services. Supply represents the minimum quantity of a particular good that producers are willing and able to sell during a specified time period.

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Law of supply: It states that as the price of good increases the quantity supplied increases and opposite is also true, as the price of good decreases the quantity supplied decreases. This makes sense intuitively since firms would likely be more willing to sell more of a good for more money rather than less of it.

Movement along the supply curve will only be caused by price; other factors will remain constant i.e. *ceteris paribus*.

**FACTORS AFFECTING SUPPLY:**

1. **The price of inputs:** In addition to the price of the product being the main factor as stated in the law of supply, the price of production inputs also plays a part. The lowest price at which a firm can sell a good without losing money is the amount of money that it costs to produce it. Producing a good or service involves taking inputs and applying a process to them to produce an output. The output is the finished good or service, and inputs are raw materials, labor, utilities and other goods. These inputs are also known as factors of production. If the price of inputs goes up, the cost of producing the good increases and therefore at each price producers need to sell their good for more money. So an increase in the price of inputs leads to a decrease in supply. Similarly, a decrease in the price of inputs leads to an increase in supply.
2. **The current state of technology:** Production technology is involved in the process part. Increases in the level of production technology can make that process more efficient with the use of technology, the operation becomes more efficient. For example, imagine that you run a basic T-shirt screen printing business out of your home. Now let's say you decide to invest in a

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workshop installed with the latest production technology. With this use of technology, the operation becomes more efficient and you are able to increase the supply of T-Shirts. If you decide to expand even further, some added technological improvements might be warranted. This further increases your ability to supply-side t-shirts sincerely it reduces labor costs.

3. **Taxation and subsidies:** Taxes imposed on goods and services are called indirect taxes e.g. sales tax. The imposition of indirect tax will bring about changes in supply. A tax on a commodity may be regarded as an increase in the costs of production. The producer then reacts by raising the price of the product through reduction in the supply of the goods i.e. the supply curve moves to the left. The vertical distance between two supply curves shows the per unit tax imposed by the government.

A subsidy, on the other hand, is a financial help provided by the government to encourage the producer to produce more output. A subsidy acts opposite to the tax i.e. it tends to reduce the cost of production causing the supply curve to shift to the right. The vertical distance between two supply curves shows the per unit subsidy by the government.

PRICE ELASTICITY OF SUPPLY (PES): It measures the responsiveness to the percentage change in quantity supplied to the percentage changes in price i.e. how the producers behave, by making wither big or small changes in the quantity which they are willing to sell, in response to changes of the good.

$$PES = \frac{\text{percentage change in quantity supplied of good X}}{\text{percentage change in price of good X}}$$

If the percentage change in quantity supplied exceeds the percentage change in price, supply is said to be elastic i.e. the producer is responsive to the change in price. In this case the value of PES will be greater than 1. On the other hand, the percentage change in quantity supplied is lesser than the percentage change in price, supply is said to be inelastic. The value in this case will be less than 1. The value of price elasticity is positive because an increase in price is likely to increase the quantity supplied to the market and vice versa.

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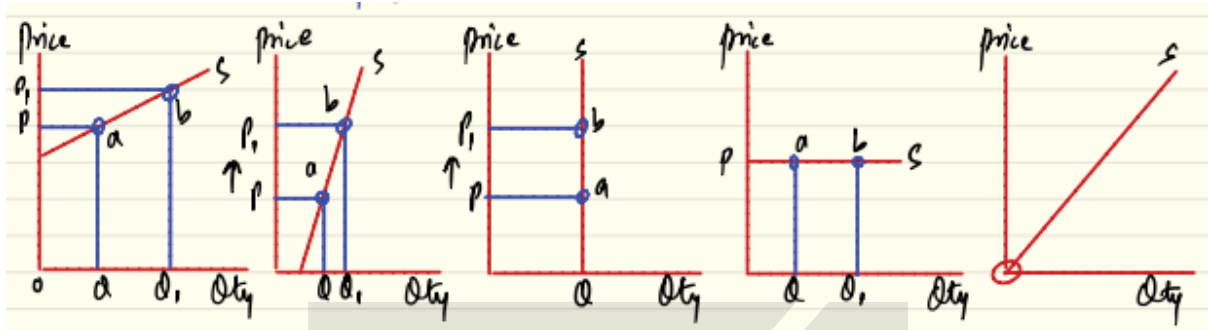
FACTORS AFFECTING PES: Elasticity of supply depends on how the producer changes the quantities of the factors of production to make changes in the quantity supplied in response to changes in price.

1. **Spare production capacity:** If there is plenty of spare capacity, a business should be able to increase its output without a rise in costs and therefore supply will be elastic in response to a change in demand. The supply of goods and services is often most elastic in a recession when there is plenty of spare labor and capital resources available to setup output as the economy recovers.
2. **Stocks of finished products and components:** If stocks of raw material and finished products are at high level then a firm is able to respond to a change in demand quickly by supplying these stocks on to the market - supply will be elastic. Conversely when stocks are low, dwindling supplies force higher and unless stocks can be replenished, supply will be inelastic in response to a change in demand.
3. **The nature of the commodity:** In the case of agricultural commodities, supply is inelastic in the short run because agricultural production depends on the harvesting period. Some agricultural commodities have inelastic supply for many years e.g. coffee plant takes several years to produce coffee beans. Thus in the short run an increase in the demand for coffee will lead to a sharp increase in its price. The supplies of products beef and milk will also be inelastic because it takes a considerable time to increase the sizes of herds of beef and dairy cattle.
4. **The ease and cost of factor substitution:** If both capital and labor resources are occupationally mobile then the elasticity of supply for a product is higher than if capital and labor cannot easily and quickly be switched.
5. **Time period:** Elasticity of supply increases with time as producers have longer time to adjust to changes in demand. Supply of most of the commodities is fixed in a particular moment in time where the PES is zero and is referred to as momentary - run. In the short run supply can be increased as a result of an increase in the price but it is usually inelastic i.e.

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PES will be less than 1. In the long run supply tends to be more elastic i.e. the value of PES will be greater than 1.



EXTREME CASES:

1. **Perfectly elastic supply:** It is a situation where a minute change in the price of a commodity leads to infinite changes in the quantity supplied and the value of PES is infinite. In this case the supply curve will be a horizontal line which shows that the producer must sell any quantity that he wants at a particular price.
2. **Perfectly inelastic supply:** It is a situation in which a proportionate change in price brings no change in quantity supplied i.e. the supply is fixed and cannot be changed in response to changes in price. In this case the value of PES will be zero and the supply curve will be a vertical line. There can be many examples of perfectly inelastic supply e.g. number of car parking spaces in a parking lot, number of seats in a concert hall or stadium, antiques. In all these cases the supply is fixed and it cannot be increased in the short run.
3. **Unitary elastic supply:** In this situation a change in price brings an equal proportionate change in quantity supplied and the value of PES will be 1 throughout the supply curve. In this case the supply curve will be a straight line starting from the origin.

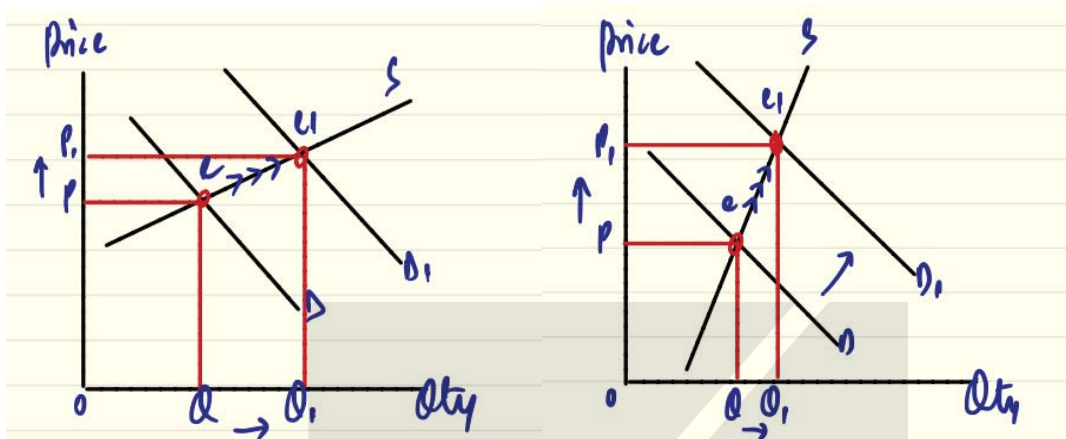
Usefulness of price elasticity of supply:

Consumers benefit from supply being elastic. This is because it means that supply is responsive to consumer demand. If demand increases, price will rise. If supply is elastic, the quantity supplied will rise by a greater percentage than the change in

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price. Sales may rise significantly without there being a large increase in price as shown in the diagram below.



Producers want their supply to be as elastic as possible. Their profits will be higher, the quicker and more fully they can adjust their supply in response to changes in demand and hence price.

Producer will benefit whether the supply is elastic or inelastic, in case of inelastic supply, with the increase in demand, a producer will be able to increase the price by greater percentage with the extension in quantity supplied by smaller percentage, and then even it will increase its total revenue.

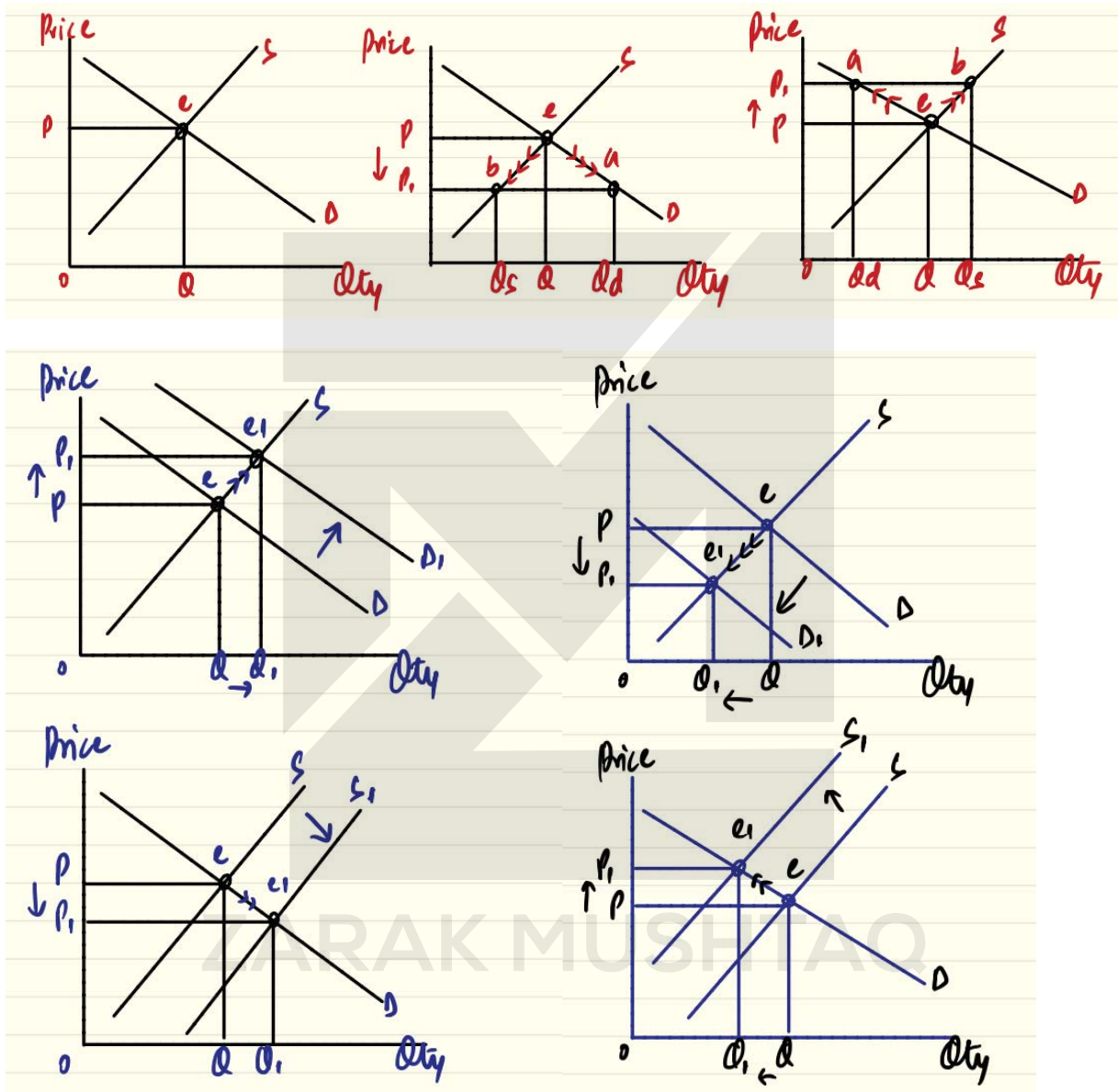
If governments want to encourage the output and consumption of a product they are likely to be more successful giving a subsidy to producers if supply is elastic. Governments use a variety of policy measures to promote flexibility in production, for example a number of governments have changed the law making it easier for firms to hire and fire labour.

MARKET EQUILIBRIUM: Equilibrium is a state of rest or balance due to the equal action of opposing forces. Equilibrium price is the price toward which the market forces of demand and supply are balanced. When a market is in equilibrium, there is no excess supply or demand. Equilibrium quantity is the amount bought and sold at the equilibrium price. The change in equilibrium position is shown below in different situations i.e. increase in demand, decrease in demand, increase in supply and decrease in supply.

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Price mechanism is where prices are determined by the forces of demand and supply where prices provide signal to the producer to allocate the resources in profit maximizing manner



MARKET SURPLUSES & SHORATAGE: A market surplus occurs where there is excess supply - that is quantity supplied is greater than quantity demanded. In this situation some producers won't be able to sell all their goods. This will induce them to lower their price to make their product more appealing. In order to stay competitive many firms will lower their prices thus lowering the market price of a

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product. In response to the lower price, consumers will increase their quantity demanded, moving the market toward an equilibrium price and quantity. In this situation, excess supply has exerted downward pressure on the price of the product.

A market shortage occurs when there is excess demand - that is quantity demanded is greater than quantity supplied. Consumers won't be able to buy as much good as they would like. Producers will raise both the price of their product and the quantity they are willing to supply. The increase in price will be too much for some consumers and they will no longer demand the product. Eventually equilibrium will be achieved.

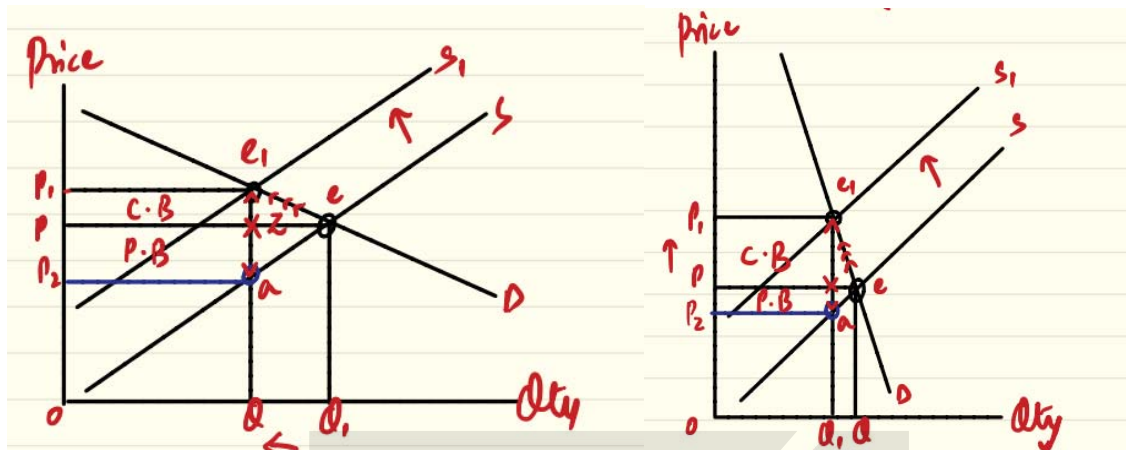
GOVERNMENT INTERVENTION IN MARKETS: Governments sometime seek to alter market outcomes. They can influence the prices at which goods are sold by taxing or subsidizing either their production or their consumption. For example if government wants to discourage the production and consumption of alcohol, it can impose heavy indirect taxes on the producers. On the other hand if the government wants to encourage the production of wheat, it will provide subsidies to the farmers.

INDIRECT TAXES: An indirect tax can be imposed i.e. specific tax; a fixed amount of tax per unit of commodity e.g. \$1 per gallon of petrol.

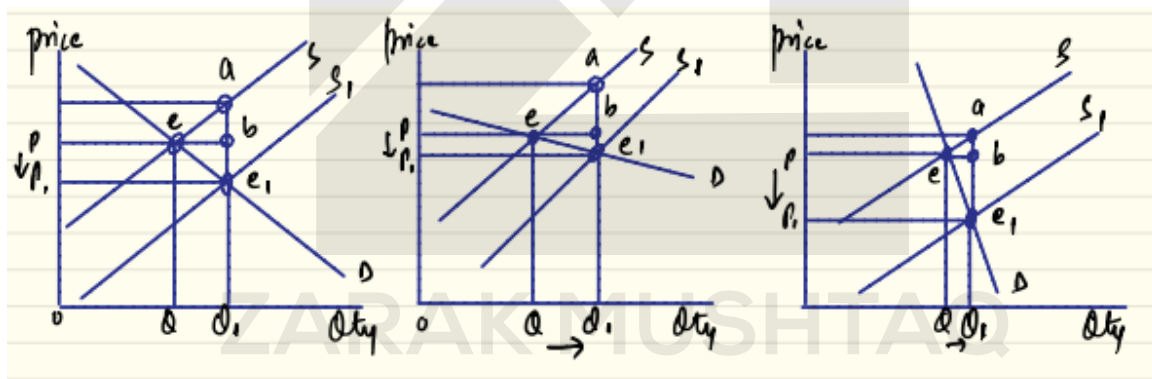
The effect of indirect tax is that it increases the cost of production of the firm, which results in decrease in the supply as shown in the diagram. Per unit tax - it is the vertical distance between two supply curves. The tax burden is shared between consumers and producers depending on the elasticity of demand.

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Subsidy: It is a grant or a cash payment given by the government to the producers to encourage them to produce more, especially in the case of basic food stuffs i.e. agricultural market. This reduces the cost of production and thus increases the supply i.e. the rightwards shift in the supply curve from S_1 to S_2 , this will result in a fall in the equilibrium price level and increase in the equilibrium quantity available in the market; hence, benefiting the consumers.



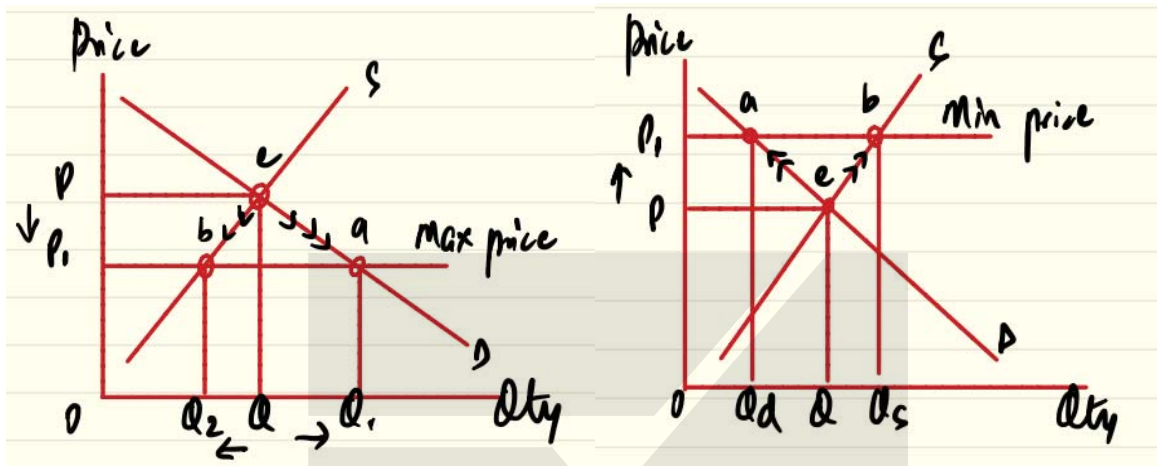
Price control: This process represents government intervention that is alternative to price mechanism.

Maximum price to be effective, it should always be imposed below the equilibrium level of output, target of the government is to reduce the price level and make it affordable for everyone but if it is imposed above the equilibrium it will be ineffective and therefore it will fall back to the equilibrium and this way price mechanism will be operational and price control will be ineffective

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It is important to note that when maximum price is imposed, though it is affordable but this results in shortage or excess demand, because when price falls, quantity demand exceeds quantity supplied.



Minimum price: To be effective it must be imposed above the equilibrium but if it is imposed below the equilibrium it will reach back to equilibrium that way price mechanism will be operational and price control will be ineffective. Once imposed effectively above the equilibrium, a rise in price contracts quantity demand but extends quantity supplied, resulting in surplus.

The target behind imposing effective price is to support the farmers as they face uncertainty in their income and make them stay in the industry, government intervenes.

ZARAK MUSHTAQ

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